

Karthikeyan S

Full Stack Engineer | React & Node.js

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Professional Summary

Full Stack Developer with 3.6 years of experience building responsive, scalable, and user-friendly web applications. Strong expertise in React, JavaScript and Node.js with a focus on performance optimization, clean architecture and reusable components. Proven ability to deliver high-quality solutions in Agile enterprise environments.

Technologies

Programming Languages: JavaScript (ES6+), TypeScript, Python

Front End Development: React.js, Next.js, Vue.js, Redux Toolkit, React Query, HTML5, CSS3, SASS, Tailwind CSS, Material UI

Back End Development: Node.js, Express.js, FastAPI, MySQL, MongoDB, REST APIs, GraphQL

Authentication & Security: JWT, OAuth2, RBAC, NextAuth.js, ZTA

Testing Libraries: Jest, React Testing Library, Cypress, Playwright

DevOps & Cloud: Git, GitHub, GitHub Actions, Jenkins (CI/CD), Apache Kafka, Redis Caching, Docker, Kubernetes, Amazon Web Services (S3, EC2, IAM, SES)

System Design & Engineering Practices: Microservices, Modular service design exposure, SOLID principles, Design Patterns, Clean architecture, Scalable system design

Experience

Systems Engineer - Tata Consultancy Services

Chennai, India | Nov 2021 - Jul 2025

Key Contributions

- Led end-to-end development of enterprise web applications using the MERN stack (MongoDB, Express.js, React 18+, Node.js), delivering scalable and maintainable solutions aligned with clean architecture principles.
- Designed and implemented secure, high-performance REST/GraphQL APIs with Node.js + Express, integrated with MongoDB Atlas and third-party services.
- Built responsive, accessible, and performance-optimized front-end applications using React with Redux Toolkit/React Query.
- Implemented JWT/OAuth2 authentication, RBAC, input validation, and encryption for financial data compliance.
- Drove CI/CD automation using GitHub Actions and Jenkins, reducing deployment failures by 35% and improving release velocity across development and production environments.
- Improved production stability by reducing post-release defects by 30% through comprehensive unit and integration testing using Jest, React Testing Library, and Cypress, consistently maintaining 90%+ test coverage.
- Collaborated closely with cross-functional agile teams across onsite and offshore locations, actively contributing to sprint planning, code reviews, and client demos to ensure timely and high-quality deliveries.
- Mentored two junior engineers, providing technical guidance, code reviews, and performance feedback that accelerated their onboarding and delivery effectiveness.

Project Details

Project 1: Guidewire - The Auto Club Group

Role: Front End Developer

Tech Stack: React.js, Redux Toolkit, React Router DOM, Material UI

Impact Highlights

- Led development of member-facing insurance portals using React and Redux Toolkit, enabling AAA members to view policies, file claims, and manage billing across web and mobile platforms.
- Built accessible and responsive UI components aligned with WCAG standards and the Guidewire Design System, improving usability across PolicyCenter and ClaimCenter applications.
- Implemented real-time quote and bind workflows using React Query, reducing quote-to-purchase time by 40% for auto and home insurance products.
- Owned frontend integration with REST and GraphQL APIs, ensuring reliable data synchronization and seamless user journeys for high-traffic insurance transactions.
- Developed a reusable component library using Storybook, standardizing UI elements such as policy cards, claims timelines, and payment widgets across multiple AAA state portals.

Project 2: Kemper Insurance - Liability

Role: Full Stack Developer & UI Automation Tester

Skills: React.js, Node.js, Express.js, MongoDB, React Testing Library, Jest

Impact Highlights

- Designed and delivered a unified multi-line insurance platform on the MERN stack, supporting Homeowners, Life, Fire (Property), Accident & Health, and Liability products for policies across 10+ states.
- Built scalable backend services using Node.js, Express and MongoDB Atlas by powering real-time quoting, policy issuance, endorsements, and renewals.
- Developed a single, responsive React 18+ front-end with Redux Toolkit and React Query, enabling dynamic product selectors and cross-line bundling logic, achieving 50% higher cross-sell conversion.
- Implemented complex rating engines for fire peril calculations, life mortality tables, A&H benefit schedules, and liability exposure calculations through configurable backend services.
- Strengthened application quality by leading UI automation testing, improving regression reliability and significantly reducing manual QA effort.

Project 3: FedEx Insurance

Role: Full Stack Developer & UI Automation Tester

Skills: React.js, Node.js, Express.js, MongoDB, React Testing Library, Jest, Tailwind CSS

Impact Highlights

- Designed and developed a global insurance & claims portal for FedEx using the MERN stack, enabling real-time cargo insurance quoting, policy issuance, and claims filing for FedEx Express, Ground, and Freight shipments.
- Built high-throughput REST and GraphQL APIs handling shipment risk scoring, premium calculations, and automated certificate generation for annual insured shipments.
- Created a responsive, white-labeled React 18 front-end with Redux Toolkit, React Query, TanStack Table, and Tailwind CSS, improving digital adoption from 18% to 60% among FedEx sales teams and customers.
- Integrated securely with FedEx core systems and third-party insurers (Chubb, AIG) using secure JWT + OAuth2 authentication and webhook-based real-time claim status synchronization.
- Implemented advanced MongoDB aggregation pipelines for insurance analytics dashboards, enabling insights into loss ratio, commodity risk trends and claims aging.
- Took on a leadership role within the team, mentoring junior developers and coordinating task delivery to ensure consistent quality and on-time releases.

Education

K. Ramakrishnan College of Technology, Trichy, Tamil Nadu

July 2016 – Oct 2020

- **Bachelor of Engineering** in Electronics and Communication
- CGPA: 7.19/10.0

Certifications

JavaScript Algorithms and Data Structures Certification, freeCodeCamp

Issued: Sept 2025

- Completed 300+ hours of coding challenges focused on JavaScript ES6, data structures, and algorithmic problem solving.

The Ultimate Node JS Course 2025: Express, MongoDB, REST API, Udemy

Issued: Oct 2025

- Completed an intensive Node.js course covering Express, MongoDB, and REST API development, gaining hands-on experience building scalable backend applications.
- Reinforced production-grade backend patterns along with international payment integration and Socket.IO bidirectional live action integration even after having the hands-on experience on Node.js backend.

Key Achievements

Star of the Month Award

Q2 2024-25

- Recognized for outstanding execution of complex UI logic and delivering seamless user experiences across high-impact enterprise applications.